

Experiment 1: Probability and Random Variables- Gaussian Distribution

Output Code

```
import numpy as np
import matplotlib.pyplot as plt

r = 35
sigma = 1
N = 100000
X = np.random.normal(r,sigma,N)
Y = np.random.normal(r,sigma,N)
plt.figure(figsize=(6,4))
plt.title("Histogram of X and Y")
plt.xlabel("Bins")
plt.ylabel("Frequency")
plt.hist(X, bins=50, label="X ")
plt.hist(Y, bins=50, label="Y ")

Xhist, Xbins = np.histogram(X, bins=50, density=True)
Yhist, Ybins = np.histogram(Y, bins=50, density=True)
Xbin_centers = 0.5 * (Xbins[:-1] + Xbins[1:])
Ybin_centers = 0.5 * (Ybins[:-1] + Ybins[1:])
plt.figure(figsize=(6,4))
plt.plot(Xbin_centers, Xhist, label="Empirical PDF X")
plt.plot(Ybin_centers, Yhist, label="Empirical PDF Y")
plt.title("Empirical PDFs of X and Y")
plt.xlabel("bins")
plt.ylabel("Relative freq")
plt.legend()
plt.grid()
plt.show()

bin_width = Xbins[1] - Xbins[0]
cdf_X = np.cumsum(Xhist * bin_width)
bin_width = Ybins[1] - Ybins[0]
```

```

cdf_Y = np.cumsum(Yhist * bin_width)
plt.figure(figsize=(6,4))
plt.plot(Xbin_centers, cdf_X, label="Empirical CDF X")
plt.plot(Ybin_centers, cdf_Y, label="Empirical CDF Y")
plt.title("Empirical CDF of X and Y")
plt.grid()
plt.legend()
plt.show()

```

```

Z = X + Y
plt.figure(figsize=(6,4))
plt.hist(Z, bins=50, density=True)
plt.title("Histogram of Z = X + Y")
plt.xlabel("Value")
plt.ylabel("Density")
plt.show()

```

```

emp_mean = np.mean(Z)
emp_var = np.var(Z)
print("Empirical Mean of Z ", emp_mean)
print("Empirical Variance of Z ", emp_var)
print("Theoretical Mean of Z ", 2*r)
print("Theoretical Variance of Z ", 2)

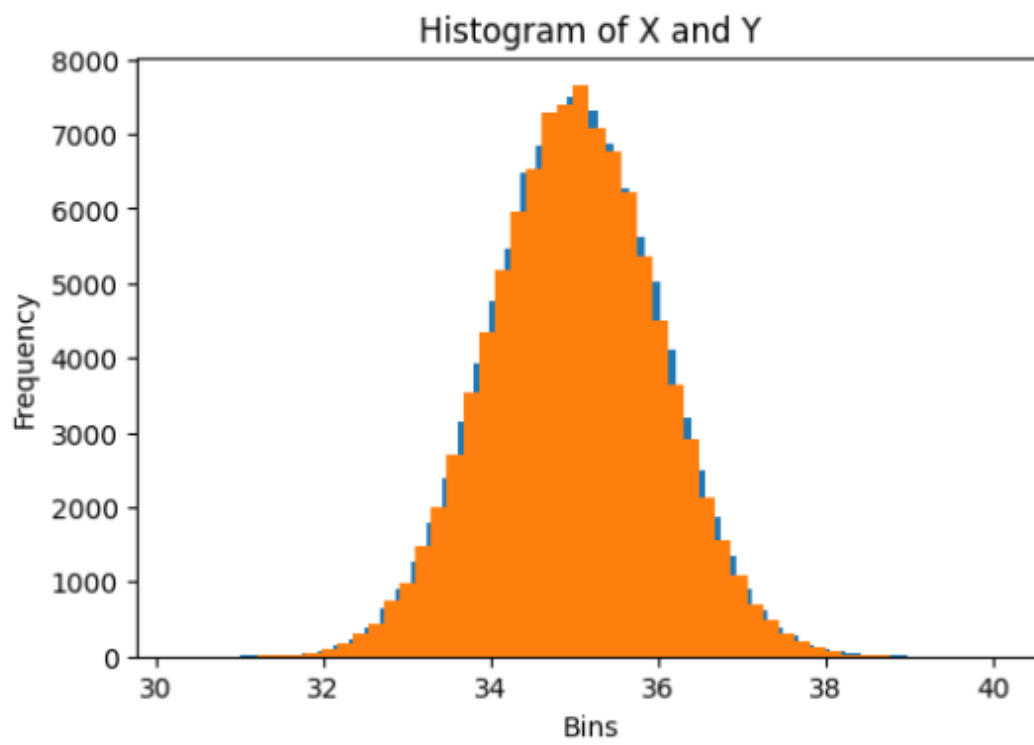
```

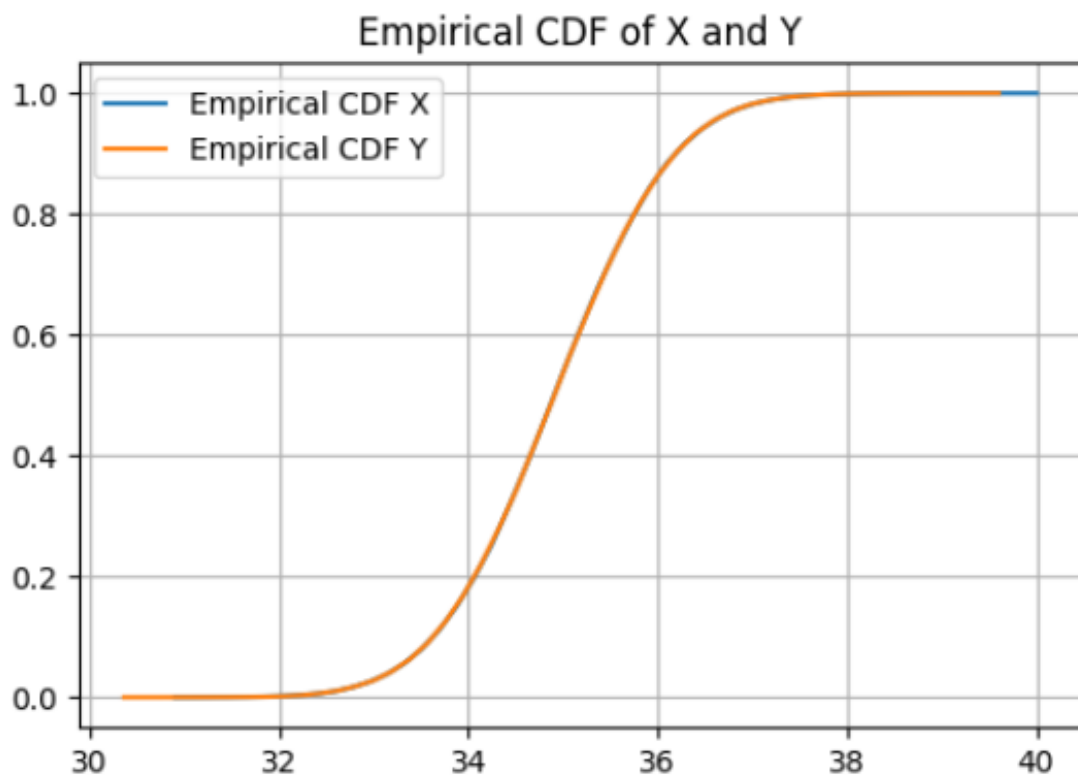
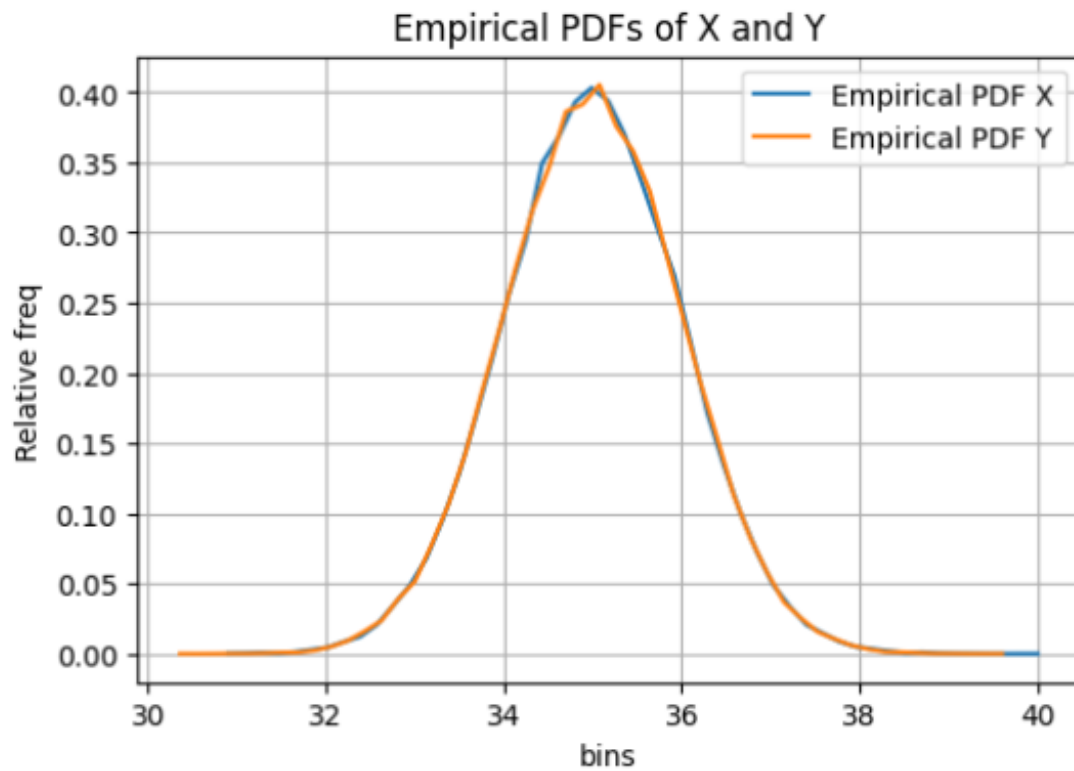
```

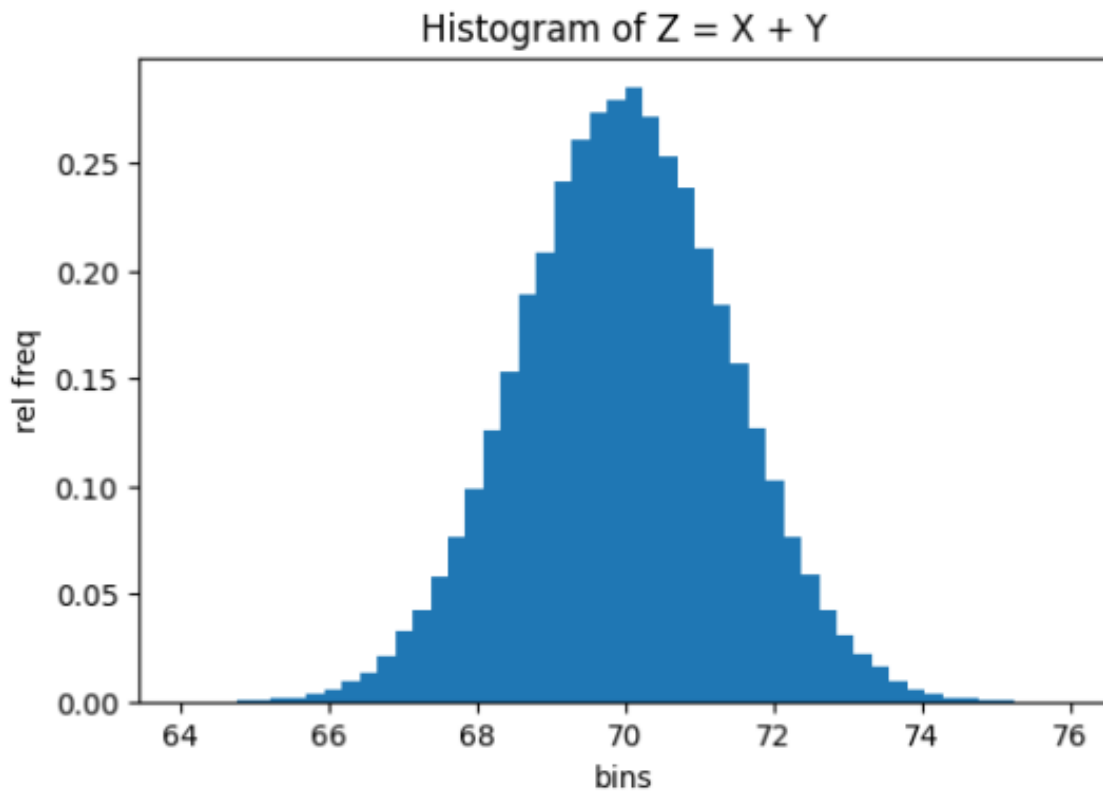
P1 = np.sum(np.abs(X - r) <= 1) / N
P2 = np.sum(np.abs(X - r) <= 2) / N
P3 = np.sum(np.abs(X - r) <= 3) / N
print("Empirical Probabilities:")
print("|X-r| <= 1:", P1)
print("|X-r| <= 2:", P2)
print("|X-r| <= 3:", P3)

```

Output







Empirical Mean of Z 70.0003551223717

Empirical Variance of Z 1.9948682707

Theoretical Mean of Z 70

Theoretical Variance of Z 2

Empirical Probabilities:

$|X-r| \leq 1$: 0.68169

$|X-r| \leq 2$: 0.95505

$|X-r| \leq 3$: 0.99747

```
plt.scatter(X, Y)
plt.title('Independent X and Y')
plt.show()

A = np.random.normal(0, 1, N)
B = 0.8*A + np.random.normal(0, 0.3, N)
plt.scatter(A, B)
plt.title('Dependent (Correlated) X and Y')
plt.show()
```

